Anesthetic Management of Asleep and Awake Craniotomy for Supratentorial Tumor Resection

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KEYWORDS

• Supratentorial tumor resection • Craniotomy • Awake craniotomy
• Neuroanesthesia • Intracranial perfusion management

KEY POINTS

- Because of compensatory changes within the fixed skull, there is often increased intracranial pressure and decreased cerebral perfusion pressure by the time supratentorial tumors are symptomatic.
- Hemodynamic monitoring and neuromonitoring can help ensure adequate cerebral perfusion; new methods of noninvasive cerebral perfusion monitoring have been developed with variable evidence of benefit.
- For patients who meet criteria, awake craniotomies remain the standard of care for tumor resection near eloquent brain regions.
- Perioperative multimodal pain control and intraoperative anesthetic adjuncts are active areas of investigation toward long-term neuroprotection and decreased mortality.

INTRODUCTION

In the United States, primary brain tumors, arising from components of the central nervous system (CNS), are among the most difficult cancers to treat, with an overall 5-year survival no greater than 35%.[1] Between 2012 and 2016, approximately 30.2% and 69.8% of CNS tumors were malignant and nonmalignant, respectively. Glioblastomas lead as the most common malignant tumor type (14.6% of all tumors and 48.3% of malignant tumors), and meningiomas remained the most common nonmalignant tumor type (37.6% of all tumors and 53.3% of nonmalignant tumors). Demographically, glioblastoma was more common in men, and meningioma was more common in women.[2] CNS tumors had an annual mortality of 4.42 per 100,000, and an average of 18,000 deaths per year, with glioblastoma having the lowest 5-year

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survival rate at 6.8%.[2] Most primary CNS tumors are supratentorial and may not be discovered until compensatory mechanisms are compromised.[1]

Understanding how anesthetics impact cerebral physiology, cerebral blood flow, brain metabolism, brain relaxation, and neurologic recovery is crucial for optimizing intraoperative and postoperative care during supratentorial craniotomies. Intraoperative goals for supratentorial tumor resection include maintaining cerebral perfusion pressure (CPP) and cerebral autoregulation, optimizing surgical access and neuromonitoring, and facilitating rapid, cooperative emergence.[3–5] Evidence-based studies increasingly expand the impact of anesthetic care beyond immediate perioperative care into both preoperative optimization and minimizing postoperative consequences. New evidence is needed for the role of neuroanesthesia in neurooncology, to prevent conversion from acute to chronic pain, and to decrease the risk of intraoperative ischemia and postoperative delirium. In this review, the authors use the most recent experimental evidence to discuss neuroanesthesia's role in improving patients' outcome and quality of life after resection of supratentorial tumors.

PATHOPHYSIOLOGY OF CEREBRAL PERFUSION, INTRACRANIAL PRESSURE, AND BLOOD-BRAIN BARRIER

The fixed space of the cranial vault enables neurophysiology and brain homeostasis to be expressed as an elegant set of theoretic equations, upon which traditional neuroanesthesia principles are based.

1. The Monro-Kellie doctrine best summarizes this relationship by stating that the additive volumes of cerebrospinal fluid (CSF), brain matter, and intracranial blood must be constant.[5] Because pressure is inversely correlated with volume in a fixed space, ICP is increased when an expanding supratentorial mass shares the cranium with the brain, cerebral blood, and CSF. Although rapidly enlarging masses (such as expanding hematomas) can cause imminent disaster within the skull and represent neurosurgical emergency, slowly growing masses are accommodated by decreasing the intracranial volume of CSF and blood before ICP elevates significantly. A gradual growth in tumor size in the classic pressure-volume curve shows little ICP change (**Fig. 1**, left of the inflection point). However, once the compensatory limit is reached (see **Fig. 1**, right of inflection point), even a small

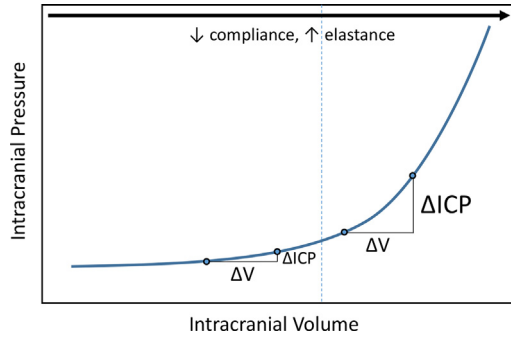


Fig. 1. Hyperbolic ICP-volume curve. Increase in intracranial volume left of the inflection point (*dotted line*) results in small ICP increase. Same increase in intracranial volume right of inflection point results in a large ICP increase. Elastance = ΔPressure/ΔVolume; Compliance = ΔVolume/Δpressure. ΔV, ΔVolume. (*Data from* Bruder N, Ravussin PA. Chapter 11: Supratentorial masses: anesthetic considerations. In: Cottrell JE, Young WL, eds. Cottrell and Young's Neuroanesthesia, 5th ed. Philadelphia: Elsevier; 2010: 184-202.)

increase in volume results in a significant elevation of the ICP.[5] Rapidly growing gliomas surrounded by vasogenic edema can double their volume within weeks, pushing the limits of compensatory mechanisms of brain compliance.

2. CPP equals the difference between mean arterial pressure (MAP) and ICP, or central venous pressure (CVP), whichever is higher. CPP = MAP − ICP (if ICP > CVP), CPP = MAP − CVP (if ICP < CVP).
3. Cushing reflex describes the physiologic response to increased ICP by elevation of systemic MAP to preserve cerebral perfusion. Hemodynamically, this manifests as systolic hypertension with widening pulse pressure, reflexive bradycardia, and decreased, irregular respirations owing to brainstem compression.[5] The presence of Cushing reflex is a neurosurgical emergency. Most strategies in neuroanesthesia revolve around maintaining CPP in the environment of increased ICP, while preventing major changes in MAP that could potentially result in brain edema, bleeding, herniation, or ischemia.
4. Cerebral autoregulation generates a stable cerebral blood flow (CBF) to meet the requirements of cerebral metabolism regardless of changing systemic blood pressure and CPP.[6,7] Indeed, the concept of neurovascular bundling demonstrates the intimate relationship between neuron metabolism and blood supply.[8] Neurons, requiring glucose for metabolic maintenance, are the first to die with prolonged anoxia or decreased energy supply. Functional autoregulation allows for wide CPP range, most commonly 70 to 90 mm Hg, traditionally 50 to 150 mm Hg, with significant variability in the lower limit between individuals and between studies.[6,9] CBF is maintained by the changes in cerebral vasomotor tone (with increasing tone corresponding to increasing cerebrovascular resistance) at the level of the cerebral arteriole (**Fig. 2**). CBF is influenced by Pa_{O_2}, Pa_{CO_2} levels, cerebral metabolic rate (CMR), systemic blood pressure, blood viscosity, and ICP.[10,11] CPP fluctuates with changes in MAP and ICP. If autoregulation is intact, tissue perfusion decreases only once MAP drops to less than 40 or 50 mm Hg.[5] Total brain ischemia is estimated to occur at CBF less than 20 mL/100 g/min.[5]

Is There an Ideal Intraoperative Cerebral Perfusion Pressure?

Optimal intraoperative MAP goal remains a matter of debate. Studies for noncraniotomy surgery have demonstrated that time spent in absolute intraoperative MAP less

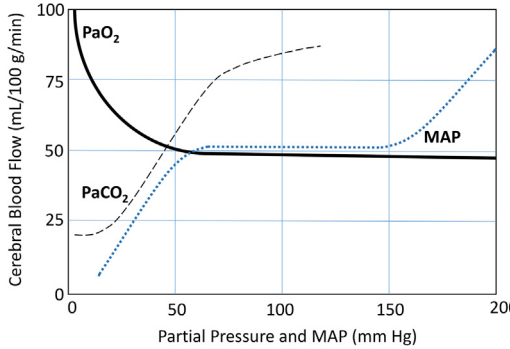


Fig. 2. Autoregulation of cerebral blood flow to MAP between 50 and 150 mm Hg. Decreasing blood flow to increased Pa_{O_2}, decreased Pa_{CO_2}, and decreasing MAP. (*Data from* Bruder N, Ravussin PA. Chapter 11: Supratentorial masses: anesthetic considerations. In: Cottrell JE, Young WL, eds. Cottrell and Young's Neuroanesthesia, 5th ed. Philadelphia: Elsevier; 2010: 184-202.)

than 55 to 65 mm Hg or a relative intraoperative MAP less than 20% of baseline has been correlated with increased myocardial infarction and acute kidney injury.[12,13] However, traumatic brain injury literature has shown that, after trauma, a significantly higher MAP is required to maintain CPP because of increased ICP.[14]

Hypertension remodels cerebral artery structure and function. During low perfusion states, vessel lumens accustomed to hypertension mechanically collapse and decrease CBP. Hypertension decreases myogenic reactivity and reduces endothelial response to vasodilatory and vasoconstricting factors, such as nitric oxide and epoxyeicosatrienoic acids.[15] Vascular responses to calcium-activated K+ channels, mechanoreceptor TRPV4, and reactive oxygen species become dysfunctional.[15] Chronic systemic hypertension impairs the renin-angiotensin-aldosterone system and likely impairs cerebral autoregulation by shifting the perfusion curve to the right.[5,16] Autoregulation is thus a complex process, not only with significant study-to-study variability but also with evidence in dramatic contrast to cerebral autoregulation dogma.[9,17]

Intraoperatively, cerebral autoregulation is further impaired by high-dose volatile anesthetics that dilate cerebral vasculature. Intravenous (IV) anesthetics tend to leave autoregulation and $Paco_2$ reactivity intact.[5] Aside from global brain dysregulation from increased ICP and chronic hypertension, the local tumor environment impairs the blood-brain barrier (BBB) by inflammation, oxidative stress, and vasoactive circulating molecules, resulting in loss of autoregulation by $Paco_2$ and MAP.[16] To maintain a continuous energy and oxygen supply that matches the delicate metabolic needs of neurons, neuroanesthesia has evolved to optimize CPP, ICP, and CBF. More evidence is needed to see if such intraoperative management offers long-term neuroprotection.

SURGICAL CONSIDERATIONS

Supratentorial brain tumors are often diagnosed after onset of symptoms, such as headaches, vision changes, sensorimotor deficits, or seizures. The symptoms arise from local tumor growth and invasion, increased local pressure, and increased pressure on the brain tissue. Neuroimaging studies, such as computed tomography MRI, will demonstrate a mass frequently surrounded by a halo of vasogenic edema.

Surgical Field Optimization

The role of the anesthesiologist is to optimize conditions for surgical exposure while preventing secondary intracranial and systemic insults (**Table 1**). Hemodynamic stability is of utmost importance during the stimulating events of induction, intubation, surgical head pinning, craniotomy, and dural opening. There may be some prolonged nonstimulating episodes as well, such as correlation of navigational probes between the patient's head and prior volumetric imaging. Brain relaxation should be achieved before dural opening to facilitate surgical access. Emergence and extubation should be ideally smooth, minimizing ICP elevation and allowing for early neurologic assessment.

Brain Relaxation, Blood-Brain Barrier, and Edema

Reducing brain volume through "brain relaxation" facilitates surgical access by neurosurgeons and decreases the amount of retraction needed.

1. The most commonly used osmodiuretic agents are mannitol and hypertonic saline. Mannitol has been used in practice for more than 50 years.[18] Dosing varies from 0.25 to 1 g/kg; 0.5 g/kg has been shown to provide sufficient surgical brain relaxation during supratentorial tumors resection for about 2 to 3 hours.[19] Furthermore, the rheologic effect of mannitol facilitates tissue perfusion. However, massive diuresis can lead

Table 1
Preventable secondary insults to compromised brain tissue

Secondary Brain Damage	Increased Intracranial Pressure	*Midline shift*: cerebral vessel tearing, hemorrhage *Herniation*: subfalcine, transtentorial, uncal, tonsillar, extracranial
	Inflammation and bleeding:	Epilepsy Vasospasm
Secondary hemodynamic changes:	Hypercapnia Hypoxemia Blood pressure changes: hypotension or hypertension Blood electrolyte changes: hypoosmolality or hyperosmolality Blood glucose changes: hypoglycemia or hyperglycemia Cardiac: decreased output, conduction changes, decreased ejection fraction Hyperthermia	

Data from Bruder N, Ravussin PA. Chapter 11: Supratentorial masses: anesthetic considerations. In: Cottrell JE, Young WL, eds. Cottrell and Young's Neuroanesthesia, 5th ed. Philadelphia: Elsevier; 2010: 184-202.

to impaired tissue perfusion and even to hypoperfusion-related organ injury. Adequate fluid resuscitation guided by goal-directed therapy may help optimize intravascular volume replacement.

Multiple studies demonstrated that hypertonic saline may achieve similar, if not better, brain relaxation without the danger of hypoperfusion.[20,21] There is also increasing evidence that hypertonic saline offers better ICP control than mannitol.[22] One metaanalysis of 9 randomized clinical trials (RCTs) confirmed the superiority of equiosmolar hypertonic saline compared with mannitol for surgical brain relaxation.[23]

Contrary to popular belief, 3% hypertonic saline can be safely administered via peripheral IV in incremental boluses of 100 to 150 mL as well as continuous infusions up to 50 mL/h[24,25] Interestingly, in a recent prospective randomized study, a large dose of 3% NaCl (5 mL/kg) was not more effective than the lower dose of 3 mL/kg for brain relaxation.[26] Adding furosemide to potentiate osmotic diuresis with hypertonic saline has marginal benefit that is not present in combination with mannitol.[27]

2. Hyperventilation is routinely used to achieve surgical brain relaxation.[10] Cerebral vasculature changes tone in response to $Paco_2$ levels. This property is independent of adrenergic stimulation despite copious vascular innervation by the autonomous nervous system.[28] There is an approximately linear relationship between CBF and acute $Paco_2$ change, 1% to 6% alteration in CBF for every 1 mm Hg of $Paco_2$ within the range of 20 to 80 mm Hg (see **Fig. 2**).[28] Extracellular pH, as well as changes in intracellular pH, plays a role in the vasomotor reactivity to $Paco_2$.[29] However, severe hyperventilation can lower jugular venous oxygen saturation and needs to be used with caution.[30] In addition, although the lower limit $Paco_2$ is yet to be established, $Paco_2$ less than 30 mm Hg can result in drastic cerebral vasoconstriction and decreased cerebral perfusion, which has correlated with global ischemia using indirect cerebral oximetry monitors.[31]

3. Optimal hemoglobin (Hgb) or hematocrit levels for blood rheology and o_2 carrying capacity are a topic outside the scope of this article. In terms of maintaining blood osmolarity and brain metabolism, normoglycemia should be maintained because both hypoglycemia and hyperglycemia are correlated with detrimental neurologic outcomes. The NICE-SUGAR trial set a reasonable goal for blood glucose at less than

180 mg/dL in critically ill patients.[32] More studies are needed to determine optimal glucose control.

Steroids for Vasogenic Edema

Systemic steroids are frequently administered for vasogenic edema in the preoperative and intraoperative period.[33] The onset of action is slow, and administration should start 24 to 48 hours before an elective surgical procedure to reduce vasogenic edema by the time of the resection.[33] Standard steroid regimen starts with a 10-mg dose of dexamethasone followed by 4 mg every 6 hours, although some studies showed good efficacy and less adverse effects with lower dose of dexamethasone at only 4 mg a day.[33] The benefits of steroid use should always be balanced against their adverse effects, such as increased delirium, agitation, immunologic depression, and endocrine changes.

Cerebrospinal Fluid Drainage

CSF production and circulation are active processes. Intracranial stasis of the CSF caused by obstruction in CSF circulation or absorption can lead to ventriculomegaly and ICP elevation. Placement of an external ventricular drain directly into the lateral ventricle allows for CSF drainage and alleviates pressure. CSF drainage through a lumbar puncture carries the risk of brain herniation with sudden pressure decrease below the foramen magnum.[5]

ANESTHETIC CONSIDERATIONS

Preanesthetic Optimization and Evaluation

In the anesthesia preoperative assessment, a thorough history and physical examination, including a detailed systems review, should be taken to elicit any concerns or comorbidities. All neurologic signs and symptoms should be noted and documented. Preoperative administration of antiepileptic and steroid therapy needs to be communicated with the surgeon. Some patients will benefit from multimodal pain control, such as initiation of gabapentin and acetaminophen, before surgery. As a precaution, providers should review advanced directives with the patient and their family.

Assessment of Risk Factors

Assessment of cardiopulmonary function is important given the fluid shifts that will occur with osmotic diuresis and blood loss, the risk of air embolism, and dysautonomia from surgical manipulation. Pulmonary comorbidities, such as chronic obstructive pulmonary disease and asthma, may interfere with hyperventilation for brain relaxation. In case of neurosurgery for brain metastases, the location and treatment of the primary tumor may have a detrimental effect on cardiopulmonary function.[34] Forty percent of brain tumors originate from primary lung tumors.[35] Chemotherapy, radiation, and prior lung surgery may decrease lung function before brain surgery.

Vascular atherosclerotic disease can predispose the patient to intraoperative myocardial ischemia as well as ischemic stroke, especially if hyperventilation results in cerebrovascular vasoconstriction. High-grade gliomas carry a clinically significant risk of venous thromboembolic event despite perioperative chemoprophylaxis.[36]

Osmotic diuresis with mannitol can lead to acute kidney injury,[37] and using hypertonic saline for brain relaxation instead of mannitol should be highly considered if renal function is compromised.[38] If any preoperative factors suggest the patient is not optimized for elective surgery, a discussion with the surgeon about additional evaluation and medical optimization may improve patient outcome.

Antiepileptics

The surgery exposes brain to surgical trauma, causing localized inflammation and stress, and increases risk of seizures. Administration of prophylactic antiepileptics (AEDs) in seizure-free patients is generally not advised.[39] Lesions with highest risk of seizures are glioneuronal tumors in the frontotemporal or insular regions.[40,41] Low-grade gliomas are surprisingly more likely to present with seizure symptoms compared with high-grade gliomas.[39] Up to 50% of patients with tumor-related seizures respond well to a single AED,[40] and studies on the use of valproic acid and levetiracetam as first-line monotherapies in tumor-related seizures suggest that preoperative AEDs should be continued throughout perioperative period.[39]

INTRAOPERATIVE MANAGEMENT

Vascular Access

Well-running, large-bore, peripheral IVs should be placed for intraoperative infusion of high-osmolarity solutions, fluids, and pressors, and to ensure good flow of IV anesthetics. Central venous access is generally not required, except for procedures with high risk of significant bleeding, pressor requirement, or air embolism.[5]

Induction of Anesthesia

The goal of induction is to avoid increases in ICP by preventing hypoxia, hypercapnia, sympathetic stimulation, and blood pressure changes.[5] The IV agents propofol and etomidate in combination with local anesthetic (lidocaine) and opioid, either in divided doses or as an infusion, are commonly used to induce anesthesia. Nondepolarizing neuromuscular blocking agents, such as rocuronium, vecuronium, and cisatracurium, are considered safe for intracranial procedures as long as they are fully metabolized or reversed prior to the start of evoked potential monitoring.[42] Beta-blockade may be considered an additional adjunct.

Airway Management

A secured airway using endotracheal intubation is preferred for supratentorial craniotomies under general anesthesia. Oropharyngeal access is likely partially obstructed by surgical positioning and head pins, and any potential kinking or dislodging of the endotracheal tube (ETT) should be preemptively prevented before surgical draping.

Fluid Management

Along with volume resuscitation using isoosmolar crystalloid solutions, transfusion threshold is a debated question in neuroanesthesia and beyond the content of this article. A restrictive transfusion threshold (Hgb < 7.4 g/dL; <8 g/dL) had no worse outcomes than liberal transfusion threshold (average Hgb > 8.7, 8–10) in a retrospective study in craniotomies.[43] Indeed, the FOCUS study provided evidence for the overall transfusion guideline to restrict transfusion threshold to Hgb less than 7.[44]

CHOICE OF ANESTHETIC FOR IMPROVED POSTOPERATIVE OUTCOMES AND NEUROPROTECTION

Current perioperative practice is moving toward enhanced recovery after surgery (ERAS) protocols. Unfortunately, because of the diversity of tumor locations and institutional protocols, Hagan and colleagues[45] stated that the “current body of evidence is insufficient to establish a standardized protocol at this time.”[4] The first RCT of use of an ERAS protocol included propofol and sufentanil for induction of anesthesia and propofol, fentanyl, and sevoflurane for maintenance. The protocol included goal-directed fluid

restriction with cardiac output guidance for hemodynamic management.[46] However, the study did not seek to measure either anesthetic depth or long-term outcomes.

Neuroprotection and Neurotoxicity

Anesthetic-related neurotoxicity has been demonstrated in vulnerable, developing brains of animal models repeatedly exposed to general anesthesia.[47] However, large human clinical trials, such as PANDA and MASK, showed that only successive, prolonged exposures before 3 years of age could possibly be associated with moderate behavioral and learning difficulties.[48,49] Currently, an active area of investigation focuses on the correlation between the anesthetic exposure (especially volatiles), and postoperative cognitive dysfunction (POCD) and delirium in the elderly.[50] Findings in human trials have been contradictory, although a major metaanalysis suggests there is weak evidence that propofol anesthesia results in less POCD than volatile anesthetic in the elderly noncardiac surgery population.[15,51–53]

In young animals, volatile anesthetics provide the most evidence for neurotoxicity. Such putative neurotoxicity may be especially relevant in the supratentorial tumor population: the BBB breakdown from tumor infiltration may render brain tissue to be as vulnerable to volatile anesthetics as the neonate brain before mature BBB formation.[51] Animal studies also showed enduring learning deficits in aged rodents after volatile anesthesia exposure (potentially from chemical toxicity).[51,52]

Delirium

There is strong motivation to decrease postoperative delirium and improve neuroprotection through perioperative protocols. Currently, the choice for total intravenous anesthesia (TIVA), volatile, or a combination of both depends on the anesthesiologist, surgeon, neuromonitoring technique, and patient factors (see **Table 1**).[3,54] Studies have shown that propofol-based anesthesia maintains lower ICP and higher CPP in comparison to anesthesia maintained by volatile agents, and propofol anesthesia is also superior in the reduction of postoperative nausea and vomiting (PONV), patient satisfaction, reduction of postoperative pain, and postanesthesia care unit length of stay, which may be factors confounding the incidence of delirium.[3,54–57] More trials are being done to assess whether intraoperative protocols lead to changes in long-term quality of life, morbidity, and mortality.[58]

Pain Control

The ongoing opioid epidemic, fueled by decades of inappropriate postoperative opioid prescription,[58] has led to increasing overdose, and death by opioid abuse.[59] There has been strong evidence that, in the craniotomy population, inadequate analgesia in the early perioperative period can lead to the conversion of acute postsurgical pain to chronic pain.[60,61] Opioids are associated with hypoventilation and hypercarbia through their sedation effect and are also tied to increases in postoperative nausea and vomiting as well as postoperative delirium.[62] Although current guidelines recommend using multimodal analgesics, opioids still remain a mainstay of postcraniotomy pain treatment.[61,63]

Anesthetic Management of Emergence

Emergence is just as critical as induction and anesthetic maintenance, and a smooth and rapid recovery without increases in MAP or ICP is critical to prevent intracranial hemorrhage and brain compression.[4] Coughing, pain, nausea, shivering, and uncontrolled blood pressure can all increase ICP, and anesthetic management should minimize these symptoms. Current RCTs on whether IV or volatile agents are the best

anesthetic for emergence have been equivocal.[4,64,65] Dexmedetomidine has become increasingly popular in nonawake craniotomy anesthetics. However, it is likely that the "manner in which anesthetic is adjusted contributes more to the quality of emergence than the specific agent used."[4] Target-controlled infusion is commonly used for conceptual modeling of infusion rate adjustments to achieve target plasma concentrations and has evidence that it may facilitate rapid emergence when used in conjunction with a bispectral index (BIS) monitor.[4] However, RCTs on the subject have been contradictory.[64,65]

ROLE OF MULTIMODAL PAIN CONTROL AND ANESTHETIC ADJUNCTS

Benzodiazepines

Administration of benzodiazepines before craniotomy is generally not recommended and should be done cautiously because of the risk of sedation leading to hypoventilation, hypercarbia, and increases in ICP.[5] Benzodiazepines can also unmask neurologic deficits and can exacerbate postoperative delirium, especially in the elderly.[66]

Intraoperative Opioids

Several studies tried and failed to provide evidence for superiority of specific anesthetic and opioid combinations during TIVA (propofol-remifentanil vs sevoflurane-sufentanil or thiopental vs propofol) in terms of time to spontaneous ventilation, time to emergence, and postoperative cognitive recovery.[64,65]

Sufentanil and fentanyl can both blunt physiologic responses to head pins and surgical stimulation, but fentanyl has a long context-sensitive half and potential for delayed awakening if poorly timed after prolonged surgery.[5] Remifentanil, with its ultrafast half-life from plasma esterase metabolism and lack of context-dependent half-life, is frequently used as an opioid adjunct for craniotomy because of its ability to promote rapid emergence. However, its rapid offset makes it the biggest culprit for opioid-induced hyperalgesia.[67]

Currently, the consensus is that multimodal analgesic adjuncts are equally important as intraoperative methods for opioid reduction, delirium prevention, and neuroprotection.[67]

Beta-Blockade

In recent literature, a 0.5 mg/kg bolus of esmolol 10 minutes before induction, followed by 0.2 mg/kg/min esmolol infusion, has been shown to decrease propofol and sevoflurane requirements, allowing for earlier extubation in an RCT.[68] Although the mechanism is not well understood, beta-2 receptor blockade may have an antinociceptive effect and decrease arousal by decreasing beta-adrenergic activation. In addition, using esmolol during a laparoscopic procedure decreases levels of inflammatory markers and may protect against surgery-induced inflammation.[69]

Nonsteroidal Anti-inflammatory Drugs

Nonsteroidal anti-inflammatory drugs (NSAIDs) can mitigate hyperalgesic response from opioid-induced hyperalgesia by regulating glutamate release, spinal cord neurotransmitters, and NMDA receptors.[67] However, nonselective NSAIDs are associated with impaired renal function and gastric ulcers, and may be especially unfavorable in craniotomies because of its platelet inhibition effects and subsequent potential for increased postoperative hematomas.[69]

Acetaminophen

Acetaminophen is a COX inhibitor without a known exact mechanism. It is thought to work centrally without NSAID risk profiles.[67] As an opioid-sparing adjunct, it is recommended to start the night before surgery and continue throughout the postoperative course. Current studies do not show an IV versus oral dosing difference.[67,70,71]

Dexmedetomidine

Dexmedetomidine is a highly selective alpha-2 agonist. In the last 10 years, its numerous benefits have led to its increased use intraoperatively. Hemodynamically, it can cause transient hypertension via alpha-1 receptor stimulation if a loading dose is administered too quickly. As a sympatholytic, it promotes hypotension and bradycardia, which can be attenuated with a slow infusion and anticholinergic agents. As an adjunctive analgesia, it can decrease intraoperative and postoperative opioid use and pain intensity, providing an excellent transitional analgesic bridge between intraoperative and postoperative care.[67] In addition, it has anxiolytic effects and decreases emergence delirium, so it can decrease benzodiazepine requirements. Other benefits include reducing the risk of postoperative bleeding from its antihypertensive properties, its antiemetic properties, and its compatibility with intraoperative neuromonitoring. Although dexmedetomidine does not increase ICP, studies in older animals found a decrease in CBF greater than the decrease in CMR. A decreased CBF velocity has also been found in humans.[72] Although this does not support the use of dexmedetomidine in the tenuous neurosurgical brain, it appeared to decrease CBF and CMR equally in later human studies, mimicking the effect of sleep.[5,73,74] Compared with remifentanil, its intraoperative use does not increase extubation time or time to orientation and allows for a fast postoperative neurocognitive examination because of the preservation of the ability to follow verbal commands.[67,75] Dexmedetomidine has been shown to interact with different CNS biochemical and circuit level pathways to decrease delirium and potentially increase neuroprotection.[52] Furthermore, animal studies described antiapoptotic and anti-inflammatory effects in the setting of anesthesia and surgery.[52]

An RCT of 80 patients receiving supratentorial tumor resection showed that intraoperative dexmedetomidine infusion decreased pain and analgesic needs as well as lessened PONV.[76] In another RCT, dexmedetomidine, compared with a remifentanil infusion, had improved pain scores postoperatively and decreased opioid utilization.[77] In addition, dexmedetomidine may offer adjunctive properties to local anesthetic by prolonging the duration of scalp block.[78] The dexmedetomidine infusion can be continued throughout the postoperative period in the intensive care unit (ICU) and help to treat pain, agitation, and delirium.[67] As a “cooperative sedative,” it provides anxiolysis and acts as an antihypertensive while preserving the ability to follow verbal commands and is particularly helpful in the context of awake craniotomies.[79]

Gabapentinoids

Pregabalin and gabapentin are GABA derivatives that bind and inhibit the alpha2delta subunit of the voltage-gated calcium channel, not the GABA receptor itself. They have been used in treatment of chronic neuropathic pain, nociceptive pain, and epilepsy.[80,81] In the neurosurgical population, they have been demonstrated to improve pain scores, decrease postoperative nausea, and decrease opioid consumption,[61,82] with strong indication for continued postoperative use.[67] One protocol showed that preoperative and postoperative administration of pregabalin (150 mg the evening

before surgery, 1.5 hours preoperatively, and postoperatively twice daily for 72 hours) reduced postoperative pain without adverse effects, such as oversedation.[82]

Local Anesthetic

Lidocaine is an amide local anesthetic that blocks voltage-gated sodium channels, but may also have off-target effects on potassium and calcium channels, neurotransmitter receptors such as NMDA and 5-HT (serotonin), and opioid receptors.[67,83] It can be used for a rapid-onset nerve block or for a local skin infiltration. Its use as an IV infusion for pain control has been described.[84] An RCT showed that a 1.5-mg/kg bolus of lidocaine after induction followed by an infusion of 2 mg/kg/h reduced postoperative pain.[4] However, it did not improve postoperative cognitive decline after supratentorial tumor surgery and may decrease seizure threshold.[85]

Dissociative Anesthetics

Ketamine is a dissociative anesthetic commonly used as an analgesic adjuvant in both intraoperative and postoperative care for noncraniotomy surgeries.[63,86] Historically regarded as having a poor neuroprotection profile because of its increase of CMR and CBF,[5] it is now being investigated in the context of improving postcraniotomy depression.[87] Brinck and colleagues[86] recommended subanesthetic dosing for pain control in order to avoid hallucinogenic properties and dose-dependent psychogenic effects.

Regional Anesthesia

There is moderate evidence that scalp blocks decrease opioid use after general anesthetic for supratentorial craniotomies.[67,78,88] Scalp block is further discussed later under Awake Craniotomy.

Ongoing Investigations for Opioid Reduction

Recently, because opioids have been linked to postoperative delirium and PONV, many protocols have focused on the concept of an "opioid-free craniotomy." Four RCTs for opioid compared with opioid-free craniotomies were reviewed in a systematic review of literature.[89] Despite heterogenous methods and weak evidence, the investigator concluded that perioperative scalp blocks, dexmedetomidine infusion, and acetaminophen were effective toward postoperative opioid-sparing by increasing time to opioid request and potentially decreasing PONV.

Additional nonpharmacologic adjuncts for opioid reduction have been explored. For instance, studies on transcutaneous electroacupuncture stimulation have suggested this may be a useful tool in reduction of postoperative opioids.[90]

INTRAOPERATIVE MONITORING: MAINTAINING ADEQUATE PERFUSION AND ASSESSING NEURAL DAMAGE

Close cardiopulmonary monitoring is essential during neurosurgical procedures to monitor and intervene on acid-base changes from brain relaxation maneuvers, assess the effect of hyperventilation, establish co_2 gradient, assess the effect of osmotic diuresis, and quantify ongoing blood loss. An arterial line is thus indicated for supratentorial craniotomies. Standard monitoring includes continuous electrocardiogram, $Etco_2$, noninvasive blood pressure and pulse oximetry, and continuous temperature monitoring, all of which has been described in multiple book chapters.[5]

Monitoring and Optimizing Anesthetic Depth

Neuroanesthesiologists must provide an anesthetic that best matches the metabolic demands of a hyperventilated, relaxed brain and does not interfere with neurophysiologic monitoring, while planning for a smooth emergence and timely recovery for a postoperative neurologic examination. Optimizing intraoperative anesthetic depth now includes increasing evidence that anesthetics may serve neuroprotective roles, improve oncologic outcomes, and improve long-term patient function. Indeed, current prerogatives involve optimizing neuroprotection, decreasing neurotoxic exposure, decreasing long-term impact on cardiopulmonary function, and improving oncologic outcome.[52,91]

Although measuring anesthetic depth is relatively standardized with age-normalized MAC values of volatile anesthetics, the prevalence of TIVA for neuromonitoring argues for the use of processed electroencephalogram (pEEG) (such as the Sedline Monitor, Masimo Corp, Irvine, CA, USA) or bi-spectral index (BIS). However, pEEG and BIS do not reliably measure anesthetic depth. Although retrospective studies have found pEEG to decrease postoperative delirium,[92] the prospective ENGAGES trial did not demonstrate the same benefit of pEEG.[93] Processed EEG may also not reduce intraoperative awareness.[94,95] However, none of these studies were done specifically for craniotomies, and most used volatile anesthetics instead of TIVA. Evidence for hysteresis of anesthetic concentrations between induction and emergence further complicates the idea of "true" anesthetic depth.[96–98] At the authors' institution, if somatosensory evoked potentials and motor evoked potentials are being monitored through scalp electrodes, EEG signals derived from electrode placement can help titrate anesthetics to avoid burst suppression.

AWAKE CRANIOTOMY

The first documented awake brain resection for the treatment of epilepsy was performed by Victor Horsley in London in 1886.[99] Awake craniotomy is now indicated for intraoperative mapping and subsequent resection of tumors, often high-grade gliomas, at or near eloquent brain tissues (particularly the language centers and primary sensorimotor cortex).[100] Other indications include epilepsy surgery, deep-brain stimulation, and most recently, cerebrovascular procedures, such as cerebral aneurysm clipping and arteriovenous malformation obliteration.[101,102]

Because of the lack of pain receptors in the brain, the awake craniotomy can be a surgical option if the patient can psychologically and cognitively tolerate a prolonged procedure, and if the surgical facility has trained the necessary multidisciplinary support staff.[54,103] The awake supratentorial tumor resection allows for a greater extent of tumor resection compared with asleep procedures, with less postoperative pain, fewer postoperative neurologic deficits, a shorter ICU and hospital stay, and improved survival.[3,4,100,104,105]

Patient Selection and Assessment

Awake craniotomy requires a motivated patient as well as a skilled multidisciplinary team, including neurosurgeon, anesthesiologist, neurophysiologist, and a neuropsychologist. The procedure has a low rate of failure regardless of American Society of Anesthesiologists class, body mass index, and comorbidities; case reports tout a successful awake craniotomy in a patient with an cardiac ejection fraction of 10%, in pregnant patients, and even in children.[4,106] Currently, the only absolute contraindication for an awake craniotomy is patient refusal. Thorough preoperative neuropsychologic assessment of the patient is required. If a patient has underlying cognitive disorders or cannot

sit still for a prolonged period, they are likely not going to tolerate an awake procedure. Likewise, patients with airway compromising factors, such as morbid obesity and obstructive sleep apnea, will likely be more challenging to manage intraoperatively.[103] In addition, if the tumor is large, vascular, or in the middle cerebral fossa floor with likelihood of high blood loss or dural and positional pain, patients may benefit from a general anesthetic.[107] No trials have been done comparing awake and asleep craniotomies in an RCT.

Choice of Sedation for Awake Craniotomy

Premedication

The consensus is to avoid medications that exacerbate somnolence, impair cognitive function, or potentially contribute to emergence delirium and confusion.[100,108] Prophylactic administration of AEDs differs with institutional practices.[100] In very anxious young patients with minimal risk of airway compromise, low-dose midazolam (1–2 mg) might be acceptable. In elderly patients, cholinergics should not be given because of the risk for confusion or delirium.[107] Acetaminophen with or without gabapentin is frequently the medication of choice at the authors' institution.

Intraoperative Management

The degree of sedation during awake craniotomy ranges from fully awake to alternating between general anesthesia and awake stages to match procedural needs. The method of asleep-awake-asleep (AAA) craniotomy has 3 distinct stages.[109] The first stage consists of scalp block, head pinning, and skull opening, when the patient is under a general anesthetic with their airway supported by a supraglottic airway device or endotracheal tube. The patient is then awake, verbally responsive, and without airway support for mapping and resection (stage 2), before being deeply sedated again with or without laryngeal mask airway (LMA) reinsertion for scalp closure and unpinning (stage 3). Alternatively, the awake craniotomy can be entirely done under monitored anesthesia care (MAC). In addition, there are reports of craniotomies and tumor resections under no chemical sedation with the aid of acupuncture or hypnosis.[110,111] Superiority of any method has not been established.[100,109] A nonrandomized retrospective analysis compared outcomes of AAA and MAC techniques at 1 institution and found that, other than a shorter operative time with the MAC technique, no differences in tumor outcomes, complications such as hypertension or seizure, or conversion to general endotracheal tube anesthesia was observed.[103]

Hemodynamic monitoring should be placed on the side ipsilateral to brain lesion in order to avoid interference with contralateral sensorimotor mapping. For either anesthetic choice, standard monitors, including electrocardiography, noninvasive blood pressure, pulse oximetry, end tidal co_2, and temperature, should be applied before initiation of sedation.[109] Some may prefer the use of intraarterial blood pressure monitoring.[109] Foley catheter placement for urine output monitoring is indicated for long surgeries or when osmotic diuresis will be used.

The success of the awake craniotomy, regardless of anesthetic choice, largely depends on the quality of the scalp block.[112,113] Blockade of 6 nerves (**Fig. 3**) is done by the neurosurgeon or anesthesiologist using a combination of local anesthetics (lidocaine, and ropivacaine or bupivacaine), with or without epinephrine, often at maximum doses.[103] Additional local anesthetic along the meningeal vessels may also decrease nerve irritation from dura opening.[107]

Awake craniotomy requires an uninterrupted high level of vigilance and attention from the multidisciplinary team that must be ready to manage complications that arise in both an AAA and an MAC technique. Decreasing agitation can be done by



Fig. 3. Targets for scalp block: (A) supraorbital nerve, (B) supratrochlear nerve, (C) zygomaticotemporal nerve, (D) auriculotemporal nerve, (E) lesser occipital nerve, (F) greater occipital nerve. (*Data from* Bruder N, Ravussin PA. Chapter 11: Supratentorial masses: anesthetic considerations. In: Cottrell JE, Young WL, eds. Cottrell and Young's Neuroanesthesia, 5th ed. Philadelphia: Elsevier; 2010: 184-202.)

continued encouragement, positional movement if needed for pressure relief, and distraction. Airway compromise is more likely for the MAC technique leading to desaturation and dangerous hypercarbia; the anesthesiologist should be ready to stop all sedative infusions and conduct airway maneuvers, such as mask ventilation with jaw thrust, placing oral or nasal airways, placing LMA, and even insertion of ETT.

Monitored Anesthesia Care

Infusion medications that can be used for MAC are dexmedetomidine, 0.2 to 0.5 μg/kg/h, propofol, 20 to 100 μg/kg/min, and remifentanil, 0.01 to 0.09 μg/kg/min. Agents and doses vary with different institutional practices.[103,109] Recent literature shows increasing use of dexmedetomidine sedation for its property of maintaining respiratory-drive while providing anxiolysis, analgesia, and antiemesis.[107] In the authors' practice, they currently use the MAC (awake-awake) method. They initiate a dexmedetomidine infusion at the high loading dose of 1 μg/kg/h for 15 to 30 minutes before arrival to the operating room with only acetaminophen premedication. Et_{CO_2} and Fi_{O_2} are provided via nasal cannula. A transport monitor continues to assess hemodynamics as the patient is brought to the operating room. The infusion is then decreased to 0.2 to 0.5 μg/kg/h. A remifentanil infusion (0.03–0.05 μg/kg/min) is added and titrated for patient and surgeon's needs. If indicated, a nasal airway can be placed when actively obstructing under moderate sedation; use of a topical vasoconstrictive agent for nasal decongestion (oxymetazoline or phenylephrine) as well as 2% or 5%

lidocaine jelly can facilitate placement.[103] Monitored limbs have to remain accessible by neurophysiologists before the scalp block and head pins are applied. Small incremental boluses of propofol (10–20 mg) or a low-dose infusion is beneficial for stimulating interventions, such as Foley catheter insertion, scalp block placement, and head pinning. Unsedated patient cooperation with the final positioning of the head is important to assure comfort and tolerance of the position. Sedation is then restarted for opening of skin and cranium. It is considered safer to only open the dura once the patient is fully awake in order to prevent herniation or surgical field compromise from wakeup agitation.[103] The patient is required to be fully cooperative for cortical mapping and lesion resection because they will have continuous neurocognitive and sensorimotor assessment via verbal, auditory, and motor cues. Mouth swabs can be given for comfort to decrease thirst. Psychologic techniques, such as conversing with the patient, verbalizing empathy, hand holding, encouragement, giving mouth swabs for thirst, warm blankets, cooling cloths, and positional shifts or cushioning, can greatly increase comfort. The direct visualization between the patient and the provider in both directions is critical.

Asleep-Awake-Asleep

If AAA with general anesthesia is preferred, increased doses of IV sedation are required: propofol (>50 μg/kg/min) along with remifentanil (0.1–0.2 μg/kg/min) and dexmedetomidine (0.2–0.7 μg/kg/h after loading dose) is frequently used; alternatively, volatile anesthetics and opioid can be used.[112] LMA is frequently preferred to an ETT because it causes less airway irritation and is therefore less likely to induce coughing and increased ICP. Once the surgeon is ready for the awake phase, a smooth transition to a cooperative and comfortable state is paramount. All anesthetics are stopped. Supplemental local anesthetic by the surgeon can be given via infiltrating between the dural leaflets around the nerves supplying dura.[103]

After tumor resection, sedation can be restarted with propofol, dexmedetomidine, and remifentanil before the LMA is reinserted for airway support.[103,114]

Neuromonitoring with Awake Craniotomy

Neuromonitoring of the awake patient during mapping and resection period is complex and multidisciplinary. Through neurophysiologic testing, hemodynamic and respiratory status should be kept constant in order to not confound testing variables with cerebral perfusion changes. Close communication of any potential problems, such as intraoperative hypertension, seizures, somnolence, agitation, respiratory compromise, bleeding, lack of brain relaxation, shivering, and nausea, is necessary.

Much feared, seizures occur with a 3% to 16% incidence during cortical and subcortical stimulation mapping.[107] The seizure and postictal state can be unsafe especially if the airway is not protected, and sudden shaking movement against head pins can also result in skull and brain trauma. Studies have also shown that intraoperative seizure activity can lead to longer hospital stays.[115] The surgeon should immediately apply cold crystalloid solution to the surgical field. If that does not abort the seizure, IV propofol in small, divided doses, and antiepileptic medications, such as ativan, levetiracetam, or fosphenytoin, may be given. Prolonged seizure as well as respiratory or hemodynamic instability is a strong indication for conversion to general anesthesia and intubation.

If the patient is especially delirious or agitated when waking up from the asleep stage, potential causes, such as pain, hypercapnia, hypoxia, or bladder distention, should be corrected.

Awake craniotomies are usually associated with a shorter ICU and hospital stay and less delirium risk.[100]

SUMMARY

Understanding how anesthetics impact cerebral physiology, cerebral blood flow, brain metabolism, brain relaxation, and neurologic recovery is crucial for optimizing both intraoperative conditions and postoperative care for supratentorial craniotomies. Using osmodiuretic agents and hyperventilation, the neuroanesthesiologist must balance brain relaxation for surgical exposure while maintaining brain perfusion to prevent primary and secondary insults. Although hemodynamic monitoring is standard of care, consciousness monitoring through processed EEGs and bispectral monitoring has not demonstrated decreased intraoperative awareness or decreased postoperative delirium. As anesthetic care extends the timescale from immediate perioperative care to preoperative optimization and postoperative quality-of-life measurements, anesthesiologists must be increasingly cognizant of how anesthetics and multimodal adjuncts affect postoperative pain control and opioid addiction risk, delirium risk, oncologic outcome, and neurocognitive decline. If tolerated by the patient, awake craniotomies at adequately staffed facilities are an excellent option for resection of brain tumors in eloquent areas and are associated with decreased hospital stays and delirium risk.

CLINICS CARE POINTS

- Brain relaxation is achieved through a combination of hyperventilation and osmodiuretic agents, and must be judicious since decreased cerebral blood flow can result in decreased cerebral perfusion pressure and global ischemia. 3% hypertonic salne can be safely administered via peripheral IV and may control intracranial pressure better than mannitol.
- A patient's baseline MAP will often shift autoregulation limits, and a MAP around 20% of the patient's awake baseline should be maintained intraoperatively to optimize brain perfusion with non-malignant ICP conditions.
- During induction and emergence, maintaining a stable MAP is paramount. If necessary, sympathetic blockade should be given to blunt stimulation, with the goal of a cooperative neurologic exam soon after extubation.
- If SSEP and MEP scalp electrodes allow EEG assessment, anesthetic depth should be titrated to EEG activity that is non-active and non-burst suppressed in order to balance intraoperative reactivity and a short emergence time.

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